# Home Lab Project Week 12, Day 1

## Task:

**Export and organize the final firewall configuration, VM configurations, network topology, Wireshark PCAP files, screenshots, scan results, vulnerability findings, and testing evidence.**

**To successfully document and conclude this home lab project, I  compiled a comprehensive, audit-ready final deployment package.** This ensures my hard work is reproducible and easily shared or reviewed.

Below is a step-by-step checklist to export and organize all assets from my VirtualBox, Kali, Ubuntu, and OPNsense environment.

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## 📂 Phase 1: Directory Structure Creation

Before exporting files, create a standardized folder structure on your host machine (or a dedicated drive) to keep assets cleanly organized.

- **`Final_Project_Lab/`**
  - 📁 `01_Configurations/` (OPNsense XML, VirtualBox OVAs)
  - 📁 `02_Network_Topology/` (Diagrams, architecture notes)
  - 📁 `03_Traffic_Captures/` (Wireshark PCAPs)
  - 📁 `04_Scan_Results/` (Nmap, OpenVAS, Nikto text/HTML reports)
  - 📁 `05_Vulnerability_Findings/` (Exploitation documentation, executive summary)
  - 📁 `06_Evidence_Screenshots/` (Proof of concept, connectivity tests)

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## ⚙️ Phase 2: Configuration & VM Exports

### 🛡️ 1. OPNsense Firewall Configuration

Do not just rely on the VM snapshot; export the actual XML configuration file.

- Log into the OPNsense WebGUI.
- Navigate to **System ➔ Configuration ➔ Backups**.
- Click **Download Configuration** to save the `.xml` file.
- *Place in:* `01_Configurations/`

### 💻 2. VirtualBox Appliance Configurations (OVA)

Exporting your VMs as OVAs creates a gold master image of your work.

- Power off all VMs (OPNsense, Kali, Ubuntu).
- In VirtualBox, go to **File ➔ Export Appliance**.
- Select your VMs (you can export them individually or together).
- Choose **Format: Open Virtualization Format 2.0 (OVA)**.
- *Place in:* `01_Configurations/`

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## 📊 Phase 3: Network Topology & Traffic Data

### 🌐 1. Network Topology Diagrams

- Export your network layout diagram (from Draw.io, Visio, or Lucidchart) as a high-resolution **PNG** and a vector **PDF**.
- Ensure it clearly marks:
  - WAN, LAN, and DMZ interfaces.
  - Subnets (e.g., `192.168.1.0/24`).
  - IP addresses of Kali, Ubuntu, and OPNsense gateways.
- *Place in:* `02_Network_Topology/`

### 🦈 2. Wireshark PCAP Files

- Open your saved Wireshark capture logs.
- Go to **File ➔ Save As** and ensure the format is `.pcap` or `.pcapng`.
- **Naming Convention:** Use descriptive names like `kali_to_ubuntu_nmap_scan.pcap` or `opnsense_blocked_dropped_traffic.pcap`.
- *Place in:* `03_Traffic_Captures/`

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## 🔍 Phase 4: Security Assessment Data

### ⚡ 1. Scan Results & Vulnerability Reports

Gather raw data from your reconnaissance phase.

- **Nmap:** Ensure you have the `.nmap` (text) or `.xml` formats of your target scans.
- **Vulnerability Scanners:** Export full **PDF** or **HTML** reports from OpenVAS, Nessus, or Nikto.
- *Place in:* `04_Scan_Results/`

### 📋 2. Vulnerability Findings Markdown

Create a quick summary document (e.g., `findings_log.md`) that acts as a mini penetration testing report.

- **Vulnerability Name** (e.g., Anonymous FTP access, Outdated Apache version).
- **Severity** (Critical, High, Medium, Low).
- **CVE Reference** (if applicable).
- **Remediation Steps** (How you configured OPNsense or patched Ubuntu to fix it).
- *Place in:* `05_Vulnerability_Findings/`

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## 📸 Phase 5: Verification & Screenshots

### 🖼️ 1. Testing Evidence Screenshots

Organize all visual evidence of your lab working successfully. Use clear naming conventions.

- `01_ping_test_kali_to_ubuntu_blocked.png`
- `02_opnsense_live_log_drop_rule.png`
- `03_kali_metasploit_success_ubuntu.png`
- `04_ubuntu_syslog_auth_failure.png`
- *Place in:* `06_Evidence_Screenshots/`

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## 🛠️ Final Step: The README Master Document

In the root directory (`Final_Project_Lab/`), create a **`README.md`** file that ties the entire package together. Include a brief executive summary, the objective of the lab, and a table of contents explaining what is inside each folder.